Seeds are shown after they have been collected and sorted as part of the Millennium Seed Bank Partnership. The project aims to store seeds of 25 percent of the world's plants by 2020.

1.8 BILLION THE NUMBER OF SEEDS CONSERVED IN THE MILLENNIUM SEED BANK

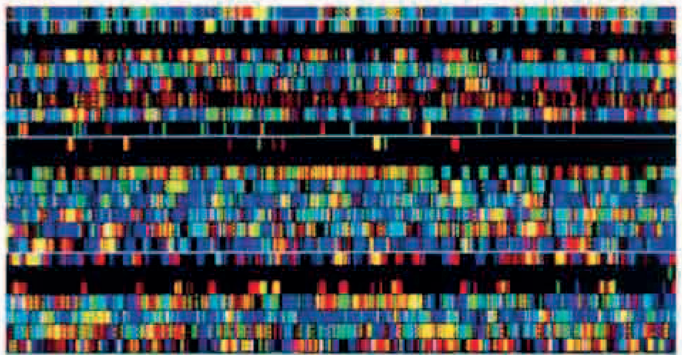
The cells inside a developing embryo have the potential to **develop into any kind of cell** (see 1981). There are similar stem cells in other tissues, but they only differentiate (develop) into a limited number of different cell types. In 1999, a team led by Italian researcher **Angelo Vescovi** (b.1962) took stem cells from mouse brains, injected them into mouse blood, and found they developed into various types of blood cell. This year, the same team took stem cells from mouse brains and found they developed into muscle cells just by being in contact with muscle cells in laboratory glassware. These experiments proved that **non-embryonic stem cells** are more flexible than was believed—a boon to stem cell research since it avoids the harvesting of embryonic stem cells, which involves the destruction of the embryo.

After a huge international effort, scientists involved in the **Human Genome Project** announced the completion of the draft sequence of the **entire human genome**. Thale cress (*Arabidopsis thaliana*), a model organism widely used in genetic experiments, became the first plant to have its genome sequence completed. The genome of another model organism, the fruit fly (*Drosophila melanogaster*), was also sequenced this year.

German biotechnologist **Ingo Potrykus** (b.1933) and German cell biologist **Peter Beyer** (b.1952) announced that they had created a **genetically modified (GM) variety of rice**. The new rice plant produced beta carotene, the precursor to vitamin A, in the edible grain. Vitamin A deficiency is commonplace in many developing countries, causing an estimated two million deaths every year, and is a major, preventable cause of blindness. The rice earned the nickname "golden rice," because of the color of the beta carotene. Biotechnology companies agreed to give the rice seeds free to farmers who made less than $10,000 per year, and the project was supported by various humanitarian organizations. However, the project has **attracted resistance** from anti-GM protesters (and anti-capitalists), who fear that multinational food companies will have an economic hold on poor farmers. Controversy delayed field trials and approval, but in 2013, golden rice seeds were given to farmers in the Philippines—with several other nations considering following suit.

In November, plant conservationists at the Royal Botanic Gardens, Kew, UK, launched the **Millennium Seed Bank Partnership**—a project involving more than 50 countries that aims to collect and preserve the seeds of tens of thousands of plants. Seeds are sorted, cleaned and dried, then stored in cold, dry conditions in large underground freezers. Climate change and changes in land use could put **many plant species at risk of extinction**, and protecting the seeds means that extinct species could be reintroduced.

In November, engineers at the Japanese car manufacturer Honda revealed the first model of their **humanoid robot ASIMO** (Advanced Step in Innovative Mobility), a popular humanlike robot that can talk, walk, and run.



HUMAN GENOME PROJECT

Officially launched in 1990, the human genome project was an international effort to determine all 3 billion DNA base pairs along the length of the human genome and identify all the 25,000 or so genes the genome contains. This knowledge holds great benefits for medical science as well as studies of human evolution and genetics in general. The project was declared complete in 2003.

> “WE ARE HERE TODAY TO CELEBRATE A **MILESTONE** ALONG A TRULY **UNPRECEDENTED VOYAGE**, THIS ONE **INTO OURSELVES.**”
>
> **Francis Collins, American geneticist**, June 26, 2000

ASIMO
The first version of Japanese car manufacturer Honda's humanoid ASIMO robot was just 4 ft (1.2 m) tall and weighed 106 lb (48 kg).

March 24 Scientists complete the sequencing of the **genome of the fruit fly**

June 26 Rough draft of the entire human genome announced by **the Human Genome Project**

October Angelo Vescovi reports that he has turned mouse **stem cells into muscle cells**

November 20 Opening of the **Millennium Seed Bank**, at Kew Gardens in the UK

November 20 Honda **debuts ASIMO**, a popular humanlike robot

December 14 The sequencing of the **genome of the model organism** thale cress is completed